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Problem

THE RESIDUAL BOTTLENECK

- Pre-Norm Transformers can suffer representation collapse and diminishing returns from added depth.
- All components read and write through one residual stream, limiting cross-layer communication bandwidth.
- Earlier dense connections use static weights shared across tokens and input roles.

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Progression

STATIC -> DYNAMIC

- Static dense aggregation learns one scalar weight per prior layer.
- Dynamic aggregation predicts a different depth-mixing vector at every sequence position from its hidden state.
- A static prior plus RMSNorm-stabilized MLP initializes the new connection weights.

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Multiway

FOUR DEPTH STREAMS

- Decouple each block input into Q, K, V and residual streams instead of reusing one tensor.
- Four depth-wise aggregators independently mix outputs from all preceding layers for each stream.
- The result acts like vertical, cross-layer four-head attention composed with horizontal token attention.

MUDDFORMER

Multiway Dynamic Dense Connections

Breaking Residual Bottlenecks in Transformers | ICML 2025

Replace the single residual highway with token-dependent dense aggregation across depth, separately routing query, key, value and residual streams into each Transformer block.

dense connections

cross-layer

dynamic routing

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Efficiency

LIGHTWEIGHT INTEGRATION

- Initialize each aggregator to select the immediately preceding layer, so training starts as a standard Transformer.
- Reallocate FFN width from 0.5x in early layers to 1.5x in late layers at fixed total parameters.
- For a 2.8B model, overhead is only 0.23% parameters and 0.4% FLOPs.

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Evidence

BETTER DEPTH UTILIZATION

- Across 405M-2.8B scales, MUDDFormer matches Transformers trained with about 1.8x-2.4x compute.
- MUDDPythia-2.8B matches Pythia-6.9B and rivals Pythia-12B in five-shot evaluation.
- Adjacent-layer cosine similarity stays lower; attention heads remain more active, especially through the value stream.

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Meaning & limits

TAKEAWAY

- Cross-layer bandwidth, not only within-layer operators, is a major scaling axis for Transformers.
- MUDD complements MoE and improves deep-narrow models; sparse connectivity trades a little perplexity for speed.
- Dense aggregation introduces extra I/O and small kernels; fused implementations and broader modality tests remain open.